DSC3032: Deep Learning 1
Project Tasks & Requirements

Your final project involves reproducing project code that has been developed by deep learning scientists to implement their novel ideas described in a scientific paper.

Here are the tasks of your project (subject to minor modifications):

1. You will prepare a tutorial to demonstrate what the paper did (data preparation, training, evaluation, inference). In order to do that, you are going to **explain the paper**, **transfer the code from GitHub to Colab** and prepare a **step-by-step code demo**.

2. Reproduce the code from the paper using Colab or any other means. Once you can clone the repo onto Colab, try running the code according to the **README** file.
   TIP: Pay attention to the arguments provided and try different settings to get started.

3. You should aim to perform **training & evaluation** in the following order of preference:
   a. **Full-training**: Train (fine-tune) the paper's model and dataset(s). Choose one representative model and dataset so that you can match the model's performance as closely as possible. You do NOT have to train all the scenarios. Note that most models would have been fine-tuned from pre-trained transformer models (trained using weights from existing models as initial parameters). In which case, you should follow their method of training as closely as possible.
   b. **Smaller dataset or model version/reduced settings**: If the dataset is too large or inaccessible, consider training on a smaller version of the dataset/model or use a subset of the main dataset or with reduced hyperparameter settings.
   c. **Different model and/or dataset**: If the dataset is not accessible or smaller versions not available, you might want to search for other benchmark datasets for the task. You could also consider a different pre-trained model (not a smaller version) that could be suitable if the paper's one is too expensive to fine-tune.
   d. If full training (fine-tuning) is not possible under all scenarios, you could **download the fine-tuned model weights and evaluate** it during demo. (Partial credit)

TIP: Look for efficient training methods (PEFT) such as LoRA which could save you time.
For any option you choose (a-d), you should show the **evaluation scores for the training and test sets**. Keep this training in one notebook, call it `GroupX-full-training.ipynb`

4. **Inference:** use any data (own or downloaded from the internet) to show visual/textual output of the model. Use another notebook, `GroupX-demo.ipynb` to show this demo live in class. The live demo notebook should contain the following:
   a. Training over 1 epoch and/or minimal hyperparameter settings. Should not take more than 2-3 minutes. If 1 epoch takes too long to run, even under minimal settings, you should explain this during your demo. You could train in advance to show that you were able to perform this, e.g. in a video.
   b. Visual output of trained model – download the weights from your fully-trained model in 3 to show how the model is inferencing on real data, e.g. performing image segmentation.

Any creativity in problem-solving to show a decent demo will be accounted for.

5. Please indicate, when presenting, the hyperparameters used by the model that you had trained, especially the number of epochs, batch size, learning rate, etc. in comparison to the settings of the paper and the evaluation metrics achieved by your model in comparison to the paper. (1 Slide)

6. Paper. You will read the paper and present the problem tackled, methods and evaluation clearly in just 10 minutes.

7. The method and evaluation are then backed up with a code demo detailing 3 and 4 in 5-8 minutes to strengthen the claim that you have understood the work done.

8. **Week 15 Presentation** (**~15 minutes**): Introduce problem, Motivations, Method(s), Datasets & Setup, Evaluation, Conclusion + Code Demo.

9. Code should be commented clearly and suitable explanations should be provided throughout instead of expecting audience/evaluators to understand the code first-hand.

10. Evaluation should be done in advance if it is time consuming, show output cell for demo showing results replicated from paper. Evaluation is extremely important!

11. If you added your own code to do certain tasks that are missing from the paper or added functionalities, please indicate this clearly in the notebook and when presenting with comment starting with `#OWN CODE`

12. Your technical merits will be awarded based on the following criteria:
    - Paper: Understanding of the problem, dataset, methods and evaluations to solve it
    - Code execution: data loading, methods (including NNs), model compilation, loss functions and optimisers, training, tweaks, evaluation.
    - Presentation: ability to articulate the ideas clearly, effectively and in engaging fashion
    - Presentation: time management
    - Group management
    - Individual contribution

13. You are recommended to use Google **Colab** for development, however, you may also use the **SKKU GPU server** for demo. You could share your files among your group using GitHub or Google Drive. Your final files should be Python notebooks (.ipynb) or Python files (.py). You could also consider collaboration platforms such as Notion, Slack, Trello, Office 365, etc. You will NOT be marked based on which tool(s) you used to collaborate, but how effectively you were able to work as a group.

14. It is important that you take initiative to CONTRIBUTE to the completion of the project. Ideally, everyone should be involved in technical contributions, even though one person could be the main technical contributor. Score will be given individually based on the technical merit AND the role you play in helping your group achieve the project goals (including communication skills and resolving conflicts).

15. Important Deadlines:
    - **End of Week 12**: Progress report 1 (Code download and working environment setup summary, paper understanding summary)
    - **End of Week 13**: Progress report 2 (Training & evaluation progress, work division, paper understanding summary)
    - **Monday/Wednesday Week 15 (June 8/10th)**: Project presentation

* Progress reports are required to be short, given in point form and to be submitted via iCampus. Any one person in the group can submit the report. Include also any work division agreed between members. These reports are not officially marked but are required to keep track of your group's progress.

* You will submit all your code, data & slides in Week 15 and submit an individual report in Week 16.